

Pediatric Organ Dysfunction Information Update Mandate (PODIUM) Contemporary Organ Dysfunction Criteria: Executive Summary

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Prior criteria for organ dysfunction in critically ill children were based mainly on expert opinion. We convened the Pediatric Organ Dysfunction Information Update Mandate (PODIUM) expert panel to summarize data characterizing single and multiple organ dysfunction and to derive contemporary criteria for pediatric organ dysfunction. The panel was composed of 88 members representing 47 institutions and 7 countries. We conducted systematic reviews of the literature to derive evidence-based criteria for single organ dysfunction for neurologic, cardiovascular, respiratory, gastrointestinal, acute liver, renal, hematologic, coagulation, endocrine, endothelial, and immune system dysfunction. We searched PubMed and Embase from January 1992 to January 2020. Study identification was accomplished using a combination of medical subject headings terms and keywords related to concepts of pediatric organ dysfunction. Electronic searches were performed by medical librarians. Studies were eligible for inclusion if the authors reported original data collected in critically ill children; evaluated performance characteristics of scoring tools or clinical assessments for organ dysfunction; and assessed a patient-centered, clinically meaningful outcome. Data were abstracted from each included study into an electronic data extraction form. Risk of bias was assessed using the Quality in Prognosis Studies tool. Consensus was achieved for a final set of 43 criteria for pediatric organ dysfunction through iterative voting and discussion. Although the PODIUM criteria for organ dysfunction were limited by available evidence and will require validation, they provide a contemporary foundation for researchers to identify and study single and multiple organ dysfunction in critically ill children.

abstract

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Pediatric critical care largely focuses on preventing, stabilizing, and hastening resolution of dysfunctional organ systems. Even in the best PICUs, recalcitrant multiple organ dysfunction syndrome (MODS) represents the most common antecedent for death.¹⁻³ Multiple investigators have ascertained that risk for mortality in the PICU is associated with number of dysfunctional organs in a dose-response fashion.^{1,4-9} More recently, risks for short- and long-term morbidity after pediatric critical illness, assessed as functional status or health-related quality of life, were strongly associated with intensity and duration of organ dysfunction.^{4,9,10}

Although the history of pediatric MODS is rich with theory and controversy, confirmation of a unifying mechanism for MODS as an underlying feature of critical illness pathophysiology remains elusive.¹¹⁻¹³ Clinical phenotypes, with individual treatment approaches, have been proposed for pediatric MODS.^{14,15} In a recent survey, parents and care providers of critically ill children reported that after survival and functional status/health-related quality of life, duration of organ dysfunction is the next most important outcome for a hypothetical interventional trial enrolling critically ill children.¹⁶

Despite its paramount importance in the practice of pediatric critical care, clinicians and researchers have relied on historical expert consensus definitions of organ dysfunctions that were derived in 2004 for the conduct of the Researching Severe Sepsis and Organ Dysfunction in Children: A Global Perspective (RESOLVE) trial of activated protein C (Xigris; Eli Lilly) for pediatric septic shock.¹⁷ Results of this consensus conference were published in 2005 as a supplement to *Pediatric Critical Care Medicine*, which is the most frequently cited reference for this journal.¹⁸ Accordingly, on March 26-27, 2015, the Eunice

Kennedy Shriver National Institute of Child Health and Human Development (NICHD) convened a group of nearly 30 experts (clinicians, basic scientists, bioengineers, and others) in Rockville, Maryland, to discuss a research agenda for pediatric MODS with an ultimate goal of improving outcomes for children who experience this common syndrome. The workshop was sponsored by the Office of Science Policy, Analysis, and Communication of the NICHD. A summary of this first Pediatric MODS Workshop was subsequently published as a supplement to *Pediatric Critical Care Medicine* in 2017.¹⁹

In addition to the development of new program announcements related to pediatric MODS (Research to Advance the Understanding and Management of MODS in Children, PAR-18-091, etc), the other immediate, clear directive that emerged during this workshop was the organization of the grassroots Pediatric Organ Dysfunction Information Update (PODIUM) Collaborative. The PODIUM Collaborative focused on the notion that to advance knowledge related to pediatric MODS, the field requires clearer, updated definitions and common data elements for MODS overall, as well as for individual organ dysfunction, particularly given the wealth of novel data that had been published on this subject over the preceding decade.

The PODIUM Collaborative addressed the key question (KQ), "What are the performance characteristics of currently used scoring tools and clinical assessments to screen for single and multiple organ dysfunction among critically ill children?" The long-term goal of the PODIUM Collaborative is to improve outcomes for children with MODS. The overall objectives of this work are to widely disseminate validated contemporary definitions of pediatric single and multiple organ dysfunction.

In this executive summary, we describe the methodology and present the final set of evidence-based pediatric organ dysfunction criteria. Additional details are provided in the accompanying articles published in this supplement of *Pediatrics*.

The feasibility of and roadmap for the PODIUM project were established in 2016 in consultation with methodologists from the Johns Hopkins Evidence-Based Practice Center (K.A.R.) and informed by targeted evidence assessment by the Scientific Resource Center, Agency for Healthcare Research and Quality, Effective Health Care Program. The methods for development of criteria characterizing organ dysfunction in critically ill children consisted of several predefined phases: (1) conduct 12 systematic reviews, including identification, assessment, and synthesis of published literature, on scoring tools and clinical assessments used for single and multiple organ dysfunction; (2) develop criteria indicating single organ dysfunction, including rationale and supporting evidence for each; and (3) undertake iterative voting for consensus building.

DEFINITIONS

Critically ill children were defined as admitted to an ICU or cared for in an emergency department or hospital ward and at risk for admission to an ICU. Pediatric age was defined as birth to <18 years, excluding critically ill premature babies (<37 weeks gestation) cared for in a NICU. Several available methods exist to identify organ dysfunction in critically ill children; therefore, we did not require a specific definition for individual organ dysfunction because we were interested in capturing a broad range of clinical, laboratory, physiologic, and imaging scoring and assessment tools used to screen for and identify organ dysfunction.

SELECTION AND ORGANIZATION OF PANEL MEMBERS

The selection of panel members was initiated by experts invited to participate in the aforementioned NICHD Pediatric MODS Workshop in 2015. Invitations were extended to experts on the basis of their record of publication on organ dysfunction topics and their leadership and participation in multicenter pediatric critical care clinical research studies during the previous 5 years. Two co-chairs were identified (M.M.B. and J.J.Z.) who were responsible for coordination of in-person meetings, conduct of educational webinars, overview of the systematic review and voting processes, and proofreading and editing of manuscripts for journal submission. Chairs for each subgroup were then identified and charged with coordination of subgroup meetings and discussions, supervision of the subgroup's progress in the conduct of its respective systematic review, evaluation of evidence for the subgroup's topic, and oversight of the subgroup's identification of criteria for organ dysfunction, accompanying rationales, any revisions needed based on voting results, and manuscript drafting. Subgroups were formed by subtopic, as follows: MODS as a unifying diagnosis; individual organ dysfunction, including neurologic, respiratory, cardiovascular, gastrointestinal, hepatic, renal, hematologic, coagulation, endocrine, immune, and endothelial; and data analysis and validation.

Ninety-two panelists were identified on the basis of their record of peer-reviewed publications on the subtopic of interest, with 4 eventually withdrawing because of time constraints. The final list of 88 panelists representing 47 institutions and 7 countries comprised the PODIUM Collaborative. Conflict of interest disclosures were sought

from all panelists before the start of the systematic reviews and again at the time of journal submission. All work was conducted voluntarily, without compensation.

SYSTEMATIC REVIEWS AND DATA SYNTHESIS

We set out to answer 2 KQs: "What are the performance characteristics of currently used scoring tools and clinical assessments to screen for (1) single and (2) multiple organ dysfunction in critically ill children?" We identified 11 subtopics for KQ1 and 1 topic for KQ2. The subtopics for KQ1 were specific to the following organ systems: neurologic, respiratory, cardiovascular, gastrointestinal, hepatic, renal, hematologic, coagulation, endocrine, immune, and endothelial. We developed Population, Intervention, Comparators, and Outcomes questions specific to each of the 11 organ systems as well as for multiple organ dysfunction as listed in table 1 of the Effective Health Care Pediatric MODS Topic Brief.²⁰ The 12 systematic reviews are reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting guidelines.²¹ The protocol for the 12 systematic reviews was registered on the International Prospective Register of Systematic Reviews (PROSPERO; CRD42018090500).

We searched PubMed and Embase from January 1992 to October 2017, with an update conducted in January 2020, by using a combination of medical subject headings terms and keywords for concepts of organ dysfunction specific to each subtopic and outcomes.²⁰ Electronic searches were conducted by medical librarians at the William H. Welch Medical Library. Search strategies, dates conducted, and number of resulting citations are detailed in Supplemental Table 1.

Studies were eligible for inclusion if the authors reported original data collected from critically ill pediatric patients (age <18 years); evaluated the performance characteristics of scoring tools or clinical assessments for organ dysfunction; and assessed an included outcome as follows: mortality (eg, PICU mortality, 28-day mortality, hospital mortality, mortality after discharge), functional outcomes/residual morbidity (eg, neurofunctional, cognitive, adaptive behavioral, depression, posttraumatic stress disorder, acute stress disorder), organ-specific outcomes/residual morbidity (eg, tracheostomy, gastric tube insertion, renal replacement therapy at discharge), outcomes related to MODS (eg, duration of new or progressive MODS, composite time to complete resolution of organ dysfunction), cost of medical care, and other patient-centered outcomes (eg, quality of life, symptom improvement, quality of dying, spillover effect of a patient's health care on loved ones).²⁰

Studies were excluded if the study population consisted of infants born preterm (<37 weeks gestation) or adults (all participants ≥18 years of age or a mixed pediatric and adult population wherein data could not be separated for patients <18 years of age). Other exclusion criteria were animal-only studies, no original data (eg, editorials, commentaries, meeting proceedings), case reports or case series with a sample size ≤10 participants, and abstract-only and non-English language publications where eligibility could not be determined.

Two independent reviewers identified studies meeting criteria for inclusion, with differences resolved by a third reviewer. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowcharts are presented for each PODIUM organ-specific systematic review.^{22–33}

TABLE 1 PODIUM: Criteria for Organ Dysfunction in Pediatric Critical Illness

Organ System	Criteria for Organ Dysfunction	Suggested Thresholds	Conditions	Severity
Neurologic	GCS	≤8	None	Not graded
	GCS-m	≤4	None	Not graded
	CAPD	≥9	None	Not graded
	EEG background attenuation and suppression; electrographic seizure activity	NA	NA in patients with history of seizures or acute neurologic injury on admission	Not graded
Respiratory	Pao ₂ /FiO ₂	≤300	If on HFNC (≥1.5 L/kg/min or ≥30 L/min), NIV, nonrebreather, or venturi; FiO ₂ ≥0.4 in all modes of support	Nonsevere
	SpO ₂ /FiO ₂	≤264	If on HFNC (≥1.5 L/kg/min or ≥30 L/min), NIV, nonrebreather, or venturi; FiO₂ ≥0.4 in all modes of support	Nonsevere
	Ventilatory failure (obstructive lung disease, eg, asthma, without oxygenation failure)	NIV	When 80% ≤ SpO₂ ≤ 97% If on HFNC (≥1.5 L/kg/min or ≥30 L/min), NIV, nonrebreather, or venturi; FiO ₂ ≥0.4 in all modes of support	Nonsevere
	OI = (FiO ₂ × mPaw × 100) / Pao ₂	Invasively ventilated ≥4 to <16	If invasively ventilated	Nonsevere
	OSI = (FiO ₂ × mPaw × 100) / SpO ₂	≥16	If invasively ventilated	Severe
		≥5 to <12.3	- If invasively ventilated - When 80% ≤ SpO₂ ≤ 97%	Nonsevere
CV ^a		≥12.3	- When 80% ≤ SpO₂ ≤ 97% - If invasively ventilated	Severe
	ECLS for any respiratory failure	NA	If invasively ventilated	Severe
	Venoarterial ECLS; temporary or durable LVAD or RVAD support	NA	None	Severe
	Cardiac arrest	NA	None	Severe
	HR	>2 SD above normal for age 0–7 d: HR >180 beats/min >1 wk to 1 m: HR >180 beats/min >1 m to <1 y: HR >180 beats/min >1 y to <6 y: HR >160 beats/min 6 y to <13 y: HR >150 beats/min 13 y to <18 y: HR >130	If present at the same time as any of the other criteria for CV organ dysfunction	Not graded
	SBP	>2 SD below normal for age 0–7 d: SBP <50 mm Hg >1 wk to 1 m: SBP <70 mm Hg >1 m to <1 y: SBP <75 mm Hg 1 y to <6 y: SBP <75 mm Hg 6 y to <13 y: SBP <80 mm Hg 13 y to <18 y: SBP <80 mm Hg	If present at the same time as any of the other criteria for CV organ dysfunction	Not graded
	Vasoactive-inotropic score ^b	≥5	If present at the same time as any of the other criteria for CV organ dysfunction	Not graded

TABLE 1 Continued

Organ System	Criteria for Organ Dysfunction	Suggested Thresholds	Conditions	Severity
Renal	Serum lactate	≥ 3 to <5 mmol/L	If present at the same time as any of the other criteria for CV organ dysfunction	Nonsevere
		≥ 5 mmol/L	If present at the same time as any of the other criteria for CV organ dysfunction	Severe
	Serum troponin I	0.6–2.0 ng/mL	If present at the same time as any of the other criteria for CV organ dysfunction	Nonsevere
		>2.0 ng/mL	If present at the same time as any of the other criteria for CV organ dysfunction	Severe
	Central venous oxygen saturation	$<70\%$	If present at the same time as any of the other criteria for CV organ dysfunction • In patients without cyanotic congenital heart disease • Ideally sampled from right atrium or pulmonary artery in a patient without intracardiac abnormalities, but proximal SVC and IVC acceptable • Whole-blood laboratory assay as standard, but consider validated continuous invasive monitoring	Not graded
	Echocardiographic estimation of left ventricular ejection fraction	30% to $<50\%$	If present at the same time as any of the other criteria for CV organ dysfunction	Nonsevere
		$<30\%$	If present at the same time as any of the other criteria for CV organ dysfunction	Severe
	Urine output ^c	<0.5 mL/kg/h for ≥ 6 h	Concomitant serum creatinine increase 1.5–1.9 times baseline ^d or ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) increase	Not graded
	Serum creatinine	<0.5 mL/kg/h for ≥ 12 h	None	Not graded
		Increase 1.5–1.9 times baseline ^d or ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) increase	Concomitant urine output ^c <0.5 mL/kg/h for ≥ 6 h	Not graded
	eGFR	Increase ≥ 2 times baseline ^d Decrease to <35 mL/min/1.73 m ²	None Excludes neonates <30 d of age	Not graded Not graded
	Initiation of RRT	NA	Initiation of RRT for any reason other than toxic ingestion or hyperammonemia	Not graded
	Fluid overload ^e	20%	Measured starting 48 h after ICU admission	Not graded

TABLE 1 Continued

Organ System	Criteria for Organ Dysfunction	Suggested Thresholds	Conditions	Severity
Gastrointestinal	Bowel ischemia	• Bowel perforation or pneumatosis intestinalis or ischemia present on gross inspection (surgical) or by plain abdominal film, CT, or MRI • Sloughing of gut	None	Severe
Hepatic ^f	AST	>100 IU/L	• Absent hemolysis or myopathy (Wilson disease is an exception as severe Coombs-negative hemolysis may be present) • Presence of liver-based coagulopathy coupled with hepatic encephalopathy^g	Not graded
	ALT	>100 IU/L	• Absent hemolysis or myopathy (Wilson disease is an exception as severe Coombs-negative hemolysis may be present) • Presence of liver-based coagulopathy coupled with hepatic encephalopathy^g	Not graded
	GGT	>100 IU/L	• Absent biliary obstruction • Presence of liver-based coagulopathy coupled with hepatic encephalopathy^g	Not graded
	Total bilirubin	>5 mg/dL (>85.5 μ mol/L)	• Absent suspected Gilbert disease • Presence of liver-based coagulopathy coupled with hepatic encephalopathy^g	Not graded
	Direct or conjugated bilirubin	>2 mg/dL (>34.2 μ mol/L)	• Absent biliary obstruction • Presence of liver-based coagulopathy coupled with hepatic encephalopathy^g	Not graded
	Liver-based coagulopathy coupled with hepatic encephalopathy	PT $\geq$ 15 s or INR $\geq$ 1.5 accompanied by clinical hepatic encephalopathy • For those with a PT $\geq$20 s or INR $\geq$2.0, hepatic encephalopathy but should be assessed, but not required • Hepatic encephalopathy is determined by age-specific grading scales²²	Presence of: • AST >100 IU/L or • ALT >100 IU/L or • GGT >100 IU/L or • Total bilirubin >5 mg/dL (>85.5 μmol/L) or • Direct or conjugated bilirubin >2 mg/dL (>34.2 μmol/L) To ensure vitamin K deficiency is not a principal component of the coagulopathy, a single dose of intravenous vitamin K (1 mg for infants up to 10 mg in adults) is administered with repeat PT/INR determined 6–8 h later	Not graded
Hematology	Platelet count ^g	<100 000 cells/ μ L	Patients without underlying hematologic or oncologic diagnoses	Not graded
		<30 000 cells/ μ L	Patients with underlying hematologic or oncologic diagnoses	Not graded
		50% decrease from baseline ^h	Patients with baseline thrombocytopenia	Not graded

TABLE 1 Continued

Organ System	Criteria for Organ Dysfunction	Suggested Thresholds	Conditions	Severity
Coagulation ^l	Leukocyte count	<3000 cells/ μ L	regardless of etiology (ie, baseline platelet count <100 000 cells/ μ L) None	Not graded
	Hemoglobin	5 to <7 g/dL	None	Nonsevere
		<5 g/dL	None	Severe
	Platelet count	<100 000 cells/ μ L	• Absent liver dysfunction • Presence of at least 1 additional coagulation dysfunction criterion	Not graded
	INR	>1.5	• Absent liver dysfunction • Presence of at least 1 additional coagulation dysfunction criterion	Not graded
	Fibrinogen	<150 mg/dL (<4.41 μ mol/L)	• Absent liver dysfunction • Presence of at least 1 additional coagulation dysfunction criterion	Not graded
Endocrine	D-dimer	>10 $\times$ the upper limit of normal ^l or above the assay's upper limit of detection if this limit is <10 $\times$ upper limit of normal	• Absent liver dysfunction • Presence of at least 1 additional coagulation dysfunction criterion	Not graded
	Blood glucose	$\geq$ 150 mg/dL ($\geq$ 8.3 mmol/L)	None	Not graded
		<50 mg/dL (<2.8 mmol/L)	None	Not graded
	Serum total thyroxine	<4.2 μ g/dL (<54 nmol/L)	NA for patients with preexisting primary or central thyroid disease	Not graded
	Serum cortisol levels before and after ACTH stimulation test	Peak <18 μ g/dL (500 nmol/L) and/or increment of <9 μ g/dL (250 nmol/L) after ACTH stimulation	• Poststimulation cortisol level should be measured at 30 min after a low-dose test and 1 h after a high-dose testing • Testing should only be considered in patients with clinical suspicion of primary adrenal insufficiency (eg, unexplained hyponatremia, hyperkalemia, hypoglycemia, hemodynamic instability)	Not graded
Immune	Peripheral absolute neutrophil count	<500 cells/ μ L	None	Not graded
	Peripheral absolute lymphocyte count	<1000 cells/ μ L	None	Not graded
	CD4 ⁺ T-lymphocyte count	<750 cells/ μ L	Age <1 y	Not graded
		<500 cells/ μ L	Age 1–5 y	Not graded
		<200 cells/ μ L	Age $\geq$ 6 y	Not graded
	CD4 ⁺ T-lymphocyte percentage of total lymphocytes	<26%	Age <1 y	Not graded
		<22%	Age 1–5 y	Not graded
		<14%	Age $\geq$ 6 y	Not graded
	Monocyte HLA-DR expression (where clinically available) ^k	< 30%	None	Not graded
	Ex vivo LPS-induced TNF- α production capacity (where clinically available) ^k	Below manufacturer provided thresholds	None	Not graded

ACTH, adrenocorticotrophic hormone; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CAPD, Cornell Assessment of Pediatric Delirium; CT, computed tomography; CV, cardiovascular; ECLS, extracorporeal life support; eGFR, estimated glomerular filtration rate; FiO₂, fraction of inspired oxygen; GCS, Glasgow Coma Score; GCS-m, Glasgow Coma Score motor response; GGT, γ -glutamyl transferase; HFNC, high-flow nasal cannula; HR, heart rate; INR, international normalized ratio; IVC, inferior vena cava; LPS, lipopolysaccharide; LVAD, left ventricular assist device; mPaw, mean airway pressure; NA, not applicable; NIV, noninvasive ventilation; OI, oxygenation index; OSI, oxygenation saturation index; Pao₂, partial pressure of oxygen, arterial; PT, prothrombin time; RRT, renal replacement therapy; RVAD, right ventricular assist device; SBP, systolic blood pressure; SpO₂, pulse oxygen saturation; SVC, superior vena cava; TNF- α , tumor necrosis factor α .

^aCriteria for CV dysfunction in patients who have CV dysfunction in the setting of critical illness, excluding patients (1) with underlying cyanotic congenital heart disease and (2) those who underwent cardiopulmonary bypass during the episode of care (ie, the ICU admission). These criteria are not intended to assess or grade impaired cardiac output or inflammatory state after cardiopulmonary bypass.

^bVasoactive inotropic score = dopamine dose ($\mu\text{g/kg/min}$) + dobutamine dose ($\mu\text{g/kg/min}$) + 100 $\times$ epinephrine dose ($\mu\text{g/kg/min}$) + 10 $\times$ milrinone dose ($\mu\text{g/kg/min}$) + 10 000 $\times$ vasopressin dose (U/kg/min) + 100 $\times$ norepinephrine dose ($\mu\text{g/kg/min}$).

^cConsider ruling out obstructive uropathy in the setting of low urine output.

^dUse the lowest serum creatinine value available in the 3 mo before admission as the baseline serum creatinine. If a previous serum creatinine is unavailable, baseline creatinine should be extrapolated from a normal eGFR for age and an appropriate estimating equation. In many critically ill children, heights are unavailable, making a height-independent equation preferential. The tables in supporting documents list estimated baseline creatinine values based on a height-independent equation and normal reference eGFR for age. These creatinine values are derived from a healthy pediatric population and have been validated in critically ill children.²⁵

^eFluid overload can be defined by input/output or wt-based calculations. For wt-based determination (kg), fluid overload = [(current wt – ICU admission wt) / ICU admission wt] $\times$ 100. For input/output-based determination, fluid overload = {[sum of daily (fluid in – fluid out)] / ICU admission wt} $\times$ 100. Use of a wt-based formula for fluid overload is preferential if wt data are available.

^fCondition that has to be met for all acute liver dysfunction criteria; no known evidence of chronic liver disease with duration of symptoms <8 wk.

^gFor the purposes of defining hematologic failure, thrombocytopenia should exist in the absence of coagulation dysfunction (ie, presence of at least 2 of the 4 PODIUM coagulation dysfunction criteria).

^hFor patients with underlying hematologic or oncologic disease and baseline thrombocytopenia, both <30 000 cells/ μL and 50% decrease from baseline criteria must be met.

ⁱWe propose that in the absence of acute liver dysfunction as defined by PODIUM, at least 2 of the 4 criteria should be present to define coagulation dysfunction. However, it should be noted that studies investigating combinations of these criteria are not available. The clinical context should be taken into account when applying these criteria in defining coagulation dysfunction. Furthermore, the proposed criteria have not been validated in children on mechanical circuits (ECLS/ventricular assist device/continuous RRT/cardio-pulmonary bypass) and, as such, may not be useful in these populations because of the effects of the circuit and associated anticoagulation therapy.

^jFoad HM, Labib JR, Metwally HG, El-Twab KM. Plasma D-dimer as a prognostic marker in ICU admitted Egyptian children with traumatic brain injury. *J Clin Diagn Res*. 2014;8(9):PC01–PC06.

^kThese tests may be clinically available outside the United States.

RISK OF BIAS ASSESSMENT, DATA ABSTRACTION, AND SYNTHESIS OF INCLUDED STUDIES

Risk of bias for included studies was assessed using the Quality in Prognosis Studies tool.³⁴ Key data elements were extracted from each study by use of an electronic form developed in REDCap³⁵ and exported into evidence tables. Graphical summaries of the risk of bias assessments and evidence tables for each subtopic are presented for each PODIUM organ-specific systematic review.^{22–33} Data synthesis was conducted by organ dysfunction assessment or scoring tool within each subtopic. Quantitative analysis was not pursued because of high heterogeneity among included studies.

DRAFTING AND DEVELOPMENT OF AGREEMENT FOR CRITERIA INDICATING ORGAN DYSFUNCTION

After completion of the systematic reviews, each subgroup proposed a set of criteria for organ dysfunction specific to its subtopic, accompanied by a rationale and supporting literature as identified through the review. In addition, suggested thresholds, any conditions that would need to be met before applying a criterion in a clinical scenario, and severity

grading were provided, as applicable. Proposed criteria, accompanying rationales, and evidence tables were then disseminated to 60 PODIUM voting members (minimum of 3 from each subgroup) through an online tool that ensured anonymity of responses (Qualtrics, Provo, UT). Each set of criteria was scored using the RAND/UCLA Appropriateness Scale, which ranges from 1 (strongly disagree) to 9 (strongly agree).³⁶ Scores of 1 to 3 represent disagreement, 4 to 6 equipoise, and 7 to 9 agreement. A comment box was optional for scores in the agreement range and mandatory for scores in the equipoise and disagreement ranges. The a priori level of agreement was set at $\geq 80\%$ of PODIUM voters who rated organ dysfunction criteria as 7 to 9. Criteria that did not reach at least 80% agreement were reviewed by the subgroup from where they originated, along with comments justifying disagreement or equipoise. The criteria were revised by the subgroup (with justification for revisions) over a period of 2 weeks. Revised criteria were re-sent to all PODIUM voters for a second round of voting. The same process was followed in a third and final round of voting. The

3 rounds of voting took place between October 18, 2019, and October 31, 2019 (round 1); November 19, 2019, and December 3, 2019 (round 2); and December 18 and 31, 2019 (round 3). Intervoting periods were used for revision of criteria that did not meet at least 80% agreement. After review of additional evidence from the January 2020 literature review update, none of the subgroups required revision of already-proposed criteria; however, risk of bias and evidence tables were updated accordingly.

In its evaluation of evidence supporting specific scoring tools or clinical assessments of organ dysfunction, each subgroup was instructed to discuss feasibility and usability of each proposed criterion (eg, “Is a laboratory test routinely obtained in the ICU?” “Are there cost limitations?” “Is the test/assessment tool invasive, resource intensive, or difficult to interpret?”).

Finally, each PODIUM subgroup identified knowledge gaps during the process of the literature review and proposed priorities for future preclinical and clinical research. Research priorities were submitted to the full PODIUM membership for ranking on a 5-tier priority scale.

PODIUM ORGAN DYSFUNCTION CRITERIA

On the basis of evidence assessed through each organ-specific systematic review of currently used scoring tools and clinical assessments to screen for single organ dysfunction, organ dysfunction criteria were proposed for all individual organ systems, with the exception of endothelial dysfunction. After the systematic review of the literature on endothelial dysfunction assessment tools, no published assessment tools or biomarkers were identified that adequately screened for or identified endothelial cell activation (ie, acquisition of new cellular functions to restore homeostasis) or dysfunction (ie, loss or inappropriate exaggeration of cellular functions worsening pathologic changes) in critically ill children.

Forty criteria were proposed initially. Eight criteria were added, and 5 were removed during voting rounds 2 and 3 on the basis of feedback from the PODIUM membership. Forty-three criteria remained after the iterative voting process described above. Median agreement scores (interquartile range) and percent agreement for each of the 3 voting rounds are detailed in Supplemental Table 2.

The organ dysfunction scoring tools and clinical assessments proposed by the PODIUM Collaborative are summarized in Table 1. The evidence tables and rationale supporting each criterion are presented in each PODIUM organ-specific systematic review.^{22–33} The top 2 priorities for future research identified by each PODIUM organ subgroup and prioritized through voting across the entire voting PODIUM membership are summarized in Table 2.

In summary, the PODIUM Collaborative was convened to review published literature on performance characteristics of

single and multiple organ dysfunction scoring tools and clinical assessments and to develop contemporary, evidence-based criteria for organ dysfunction in critically ill children. These goals were achieved by conducting systematic reviews of the literature for single and multiple organ dysfunction scoring tools and assessments and by building consensus for the resulting criteria. The PODIUM criteria for organ dysfunction are meant to serve as a foundation upon which researchers can further validate, refine, and combine criteria to accurately identify patients with single or multiple organ dysfunction; to identify patterns of organ dysfunction combinations and temporal trends that constitute unique phenotypes associated with worse outcomes; and to serve either as entry criteria or as outcome measures for clinical trials, depending on the nature and scope of the interventions tested.

The PODIUM project has several strengths. It is the first large-scale summary of existing evidence related to performance characteristics of scoring tools and assessments for organ dysfunction in critically ill children. Previously proposed criteria have been based on expert opinion, with potential bias inadvertently introduced by panel members. All systematic reviews conducted for PODIUM were rigorous, transparent, and fully reproducible; the search strategy is published along with this executive summary (Supplemental Information), thus facilitating regular updates as new evidence and novel tests for organ dysfunction emerge. Special emphasis was placed on developing organ dysfunction criteria that are strongly supported by published studies and not by expert opinion. In addition, whenever possible, we

took into consideration issues of (1) feasibility (ie, tests or clinical assessments that can be conducted routinely in most critically ill children), (2) safety (ie, preference for noninvasive or minimally invasive tests over invasive tests, even if the latter have better performance characteristics), (3) equity (ie, tests or clinical assessments that can be performed in ICUs, including those with limited resources), (4) limitations for timing of assessment (ie, generalizable to the entire ICU or hospital stay versus studied only on specific days, such as day of admission to the ICU), (5) barriers to accurate organ dysfunction assessments (ie, difficult-to-interpret tests or tests requiring a high level of training and specialization), and (6) operationalization (ie, tests or clinical assessments routinely recorded in electronic medical records that facilitate future development of clinical decision support tools).

Limitations are related to available data as well as the PODIUM process. Extreme heterogeneity exists in the categorization of various patient populations among pediatric critical care studies and in definitions of common data elements, including “basic” data elements (eg, age categories). This heterogeneity rendered quantitative evaluation of performance characteristics of individual scoring and assessment tools in the form of meta-analyses inappropriate. Although the PODIUM Collaborative emphasized diversity of institutions and diversity of age and gender among participating members, initial membership was dictated by participation in the 2015 NICHD symposium. We acknowledge that although membership was broadened and included members from 7 countries, it is still primarily representative of academic North American pediatric ICUs.

TABLE 2 PODIUM: Research Priorities for the Study of Organ Dysfunction in Critically Ill Children

PODIUM: Research Priorities
Develop and validate tools that use routine clinical data from the electronic health record that allow for automated and longitudinal calculation of scores to be made available for “real-time” clinical assessment.
Develop scores to predict—rather than diagnose or describe—organ dysfunction.
Identify trajectories or early warning signs of cardiovascular dysfunction in critical illness for prediction of clinical deterioration to cardiopulmonary arrest or institution of mechanical circulatory support. Can these be used to target early intervention in this high-risk population?
Validate urinary biomarkers of AKI/renal dysfunction: (1) appropriate thresholds in children, in particular noncardiac populations, (2) use of biomarkers to derive and target MODS-AKI phenotypes, and (3) development of a clinical renal function panel.
Compare the epidemiology and outcomes of MODS as a syndrome versus coexisting, but pathobiologically distinct, concurrent multiple organ dysfunctions.
Identify biomarkers (eg, proteomic and/or transcriptomic signatures) of immune system dysregulation in critically ill children to develop high-throughput, rapid-turnaround tests for these biomarkers and to move them to clinical laboratory and/or the bedside for the diagnosis and management of immune system failure in critically ill children.
Validate objective scoring systems for neurologic dysfunction in pediatric MODS that can be used longitudinally to detect (1) patients at risk for neurologic injury, (2) progression of injury, and (3) resolution/repair of the injury.
Ascertain impact of bundled care for AKI (eg, use of balanced fluids, nephrotoxin avoidance, diuretics).
Identify and prognosticate according to existing and emerging technology (somatic and cerebral NIRS, analyses of cardiac index, echocardiographic parameters) in the assessment of cardiovascular dysfunction in critical illness. Can any be associated with improvement in clinical status with therapy?
Correlate biomarkers to physiologic function and measured clinical parameters. Unbiased, large-scale analysis, so-called “-omics” approaches, should be used to monitor multiple variables simultaneously and provide novel insights into disease pathology.
Evaluate host-microbial interactions in the gastrointestinal tract.
Define coagulation dysfunction using different combinations of the laboratory tests included in the proposed criteria in children off and on mechanical circuits (high research priority).
Expand the definition of respiratory failure in MODS beyond gas exchange. Oxygenation and ventilation are nonspecific, affected by cardiac function, and do not address pathophysiology. Biomarkers of lung epithelial and endothelial disruption may provide additional structural and pathophysiologic information. Can biomarkers of pulmonary damage improve the definition of respiratory MODS?
Investigate the performance of von Willebrand factor, antithrombin, thrombomodulin, mean platelet volume, thromboelastography/thromboelastometry to further define coagulation dysfunction.
Explore correlation with critical illness outcomes and consider implications for clinical research for markers of adrenal axis function at the cellular level.
Facilitate high-throughput, rapid-turnaround tests of leukocyte function (eg, HLA-DR expression, cytokine production capacity) to the clinical laboratory and/or the bedside for clinical use for the diagnosis and management of immune system failure in critically ill children.
Determine if effective minute ventilation via invasive or noninvasive measures improve the definition of respiratory MODS.
Consider whether the definition of hematologic failure should include abnormal function in addition to abnormal quantity of cells/cellular components.
Identify mechanistic links between neurologic dysfunction and other organ dysfunctions (eg, exosomes released from liver triggering neurologic involvement, sepsis pathophysiology mechanisms, and neurologic dysfunction).
Develop a more fundamental understanding of how endothelial cells from various organs and vascular segments differentially respond to stimuli associated with critical illness that is focused on human cell models with defined properties of specific vascular segments or organs.
Develop a reliable clinical score and/or biomarkers of feeding intolerance.
Explore correlation of copeptin (an indirect measure of ADH/vasopressin concentrations) with critical illness outcomes.
Characterize acute-on-chronic liver failure in children to provide a foundation to develop consensus guidelines.
Determine if red cell distribution width is a clinically relevant biomarker of hematologic failure.
Characterize disseminated intravascular coagulation in the setting of acute liver failure.

In descending order of priority on the basis of PODIUM membership ranking. ADH, antidiuretic hormone; AKI, acute kidney injury; NIRS, near-infrared spectroscopy.

CONCLUSIONS

The PODIUM criteria for organ dysfunction provide a foundation for clinicians and researchers to diagnose and study single and multiple organ dysfunction in critically ill children. These criteria will require further validation and

refinement followed by implementation in the clinical environment with the use of bioinformatics tools. The PODIUM process is transparent and reproducible to facilitate serial updates as new evidence and novel criteria for organ dysfunction emerge.

ABBREVIATIONS

- KQ: key question
- MODS: multiple organ dysfunction syndrome
- NICHHD: Eunice Kennedy Shriver National Institute of Child Health and Human Development
- PODIUM: Pediatric Organ Dysfunction Information Update Mandate

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